TECHNICAL REPORT

Soil Impoverishment

A Little-Known Technique Holds Potential for Establishing Prairie

by John P. Morgan

Depleting the soil of nitrogen gives prairie plants a chance to demonstrate their tenacity.

As part of the Tall Grass Prairie Restoration Project in Manitoba, Canada, described elsewhere in this issue, we initiated an experiment to test a little-known technique called soil impoverishment for its utility in enhancing the restoration process. Soil impoverishment is based on the innate ability of prairie plants to tolerate and flourish in soils with low levels of nitrogen.

Soil impoverishment, or reverse fertilization, involves the removal of nutrients from the soil. It is a technique regularly used in the forestry industry to promote the growth of certain species over others (Wilson, 1992). It is most often done by introducing large amounts of organic matter to the soil. Microbial decomposition of this organic matter temporarily monopolizes large amounts of nitrogen, making it unavailable to plants.

The results of our field work and recently reviewed information gave us a suspicion that soil impoverishment may significantly enhance prairie restoration. Incorporation of oat hulls as a carbon source lowered soil nitrate and weed levels in an Ontario experiment with legumes (REAP Canada, 1991). Native warm-season grass species have an innate ability to use nitrogen more efficiently than cool-season exotics such as quack grass (*Agropyron repens*) and Kentucky bluegrass (*Poa pratensis*), and can effectively outcompete them in low nitrogen conditions (Wedin and Tilman, 1992).

Three soil impoverishment sub-plots were established in 1992 using a mixture of sawdust and sugar (10 kg (22 lbs) sugar mixed with 80 litres (21 Imp. gal) sawdust raked into a 5-m^2 (6-yd^2) area). The areas were then seeded with a Truax Model 88 flex drill to a mix of 29 local ecotype native grasses and forbs.

The soil impoverization plots on one of our three research sites in 1992 showed a significant reduction in weed growth with no decrease in the growth of native prairie species. The effect was so dramatic that drill rows of prairie species were evident throughout the plot while weed growth, which was abundant on the rest of the site, virtually stopped at sub-plot boundaries. Very few weeds came up on the impoverished sub-plot. This growth pattern lasted for more than one year until the early part of the 1993 growing season.

The other two sub-plots were on a different soil type that was much higher in initial nitrogen levels and showed no effect on the prairie plants or weeds. We hypothesized that not enough organic matter was added to these two plots to make a measurable difference in soil nitrogen levels. Initial soil nitrogen levels were much higher on these two sub-plots than on the first.

We attempted three replicates of the experiment on a black chernozemic silty clay loam in 1993. These experiments involved rototilling 12 kg (26 lbs) of barley straw as a carbon source into 5-m^2 (6-yd^2) plots and then drilling big bluestem (*Andropogon gerardi*) at a rate of 12 kg/ha (13 lbs/acre). Flooding destroyed these sub-plots in 1993, but we plan further research in 1994.

Weed control is a primary concern in all prairie restorations, making the concept of soil nitrogen manipulation to inhibit weedy plant growth one that merits further research. Traditional herbicides are of limited value in prairie restoration work because of their harmful side-effects on the prairie plants being established. But weed growth often has a significant impact upon the successful establishment of a prairie planting. In particular, in the productive black chernozemic soils that cover much of North America's tallgrass prairie region, weed growth can be nearly astronomical, severely limiting the success of slower-growing native species.

If soil impoverishment is proven to be useful in overcoming weed competition, it would be a major advance in the field of prairie restoration. We feel that our results, although preliminary, may encourage others in North America to experiment with this technique, which may in turn help to remove a significant stumbling block to restoring our native prairie heritage. We are happy to correspond with anyone interested in this method to help them set up a suitable experimental test of the concept. We also would like to hear from anyone who may have tried this technique and subsequently review their results.

For anyone planning such an experiment, the following is a suggested outline of its implementation:

1. Choose a potential prairie restoration site with a known history of non-native weed problems.
2. Have a representative sample of the soils from the site tested by a soils lab, paying particular attention to available soil nitrogen levels.
3. Lay out several experimental plots and control plots, all the same size and representative of the area as a whole.
4. With soil analysis in hand and the area of each plot known, consult with soil scientists to determine the amount of organic matter needed to add to each of the experimental plots and thus tie up available soil nitrogen for at least one growing season.
5. Apply the recommended amounts of a soil-impoverizing substance to each experimental plot, then rake, cultivate or rototill in to incorporate it. The impoverizing substance could be sugar/sawdust, oat hulls, rice hulls, peat moss, clean straw, or any other weed-free organic matter available.
6. Seed the experimental and control plots with the same native prairie species at the same rate and time, and in the same manner.
7. Monitor each plot for its relative densities and cover of both native species and weeds for at least one growing season.

The soil scientists, agrologists, and prairie specialists that we have run this idea by have all agreed that the soil impoverishment theory merits a closer look. Exploiting the native prairie species' ability to tolerate low nitrogen levels while also taking advantage of most weed species' need for abundant nitrogen may well be a productive area for future restoration research.

References

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